1.10 Identifying the Graphs of Linear, Quadratic, Cubic and Reciprocal Functions

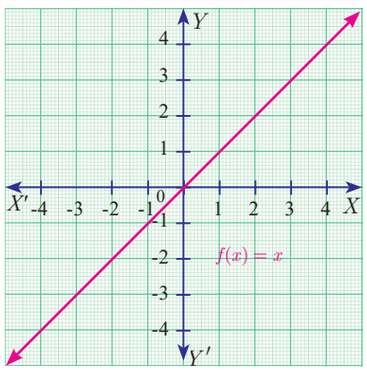
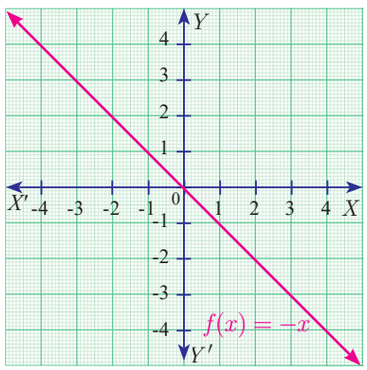
Graphs provide visualization of curves and functions. Hence, graphs help a lot in understanding the concepts in a much efficient way.

In this section, we will be discussing about the identification of some of the functions through their graphs. In particular, we discuss graphs of Linear, Quadratic, Cubic and Reciprocal functions.

1.10.1 Linear Function

A function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = mx + c$, $m \neq 0$ is called a linear function. Geometrically this represents a straight line in the graph.

Some Specific Linear Functions and their graphs are given below.

No.	Function	Domain and Definition	Graph
1	The identity function	$f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x$	 Fig. 1.44
2	Additive inverse function	$f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = -x$	 Fig. 1.45

1.10.2 Modulus or Absolute valued Function

$f : \mathbb{R} \to [0, \infty)$ defined by $f(x) = |x|$

$$= \begin{cases} x; & x \geq 0 \\ -x; & x < 0 \end{cases}$$

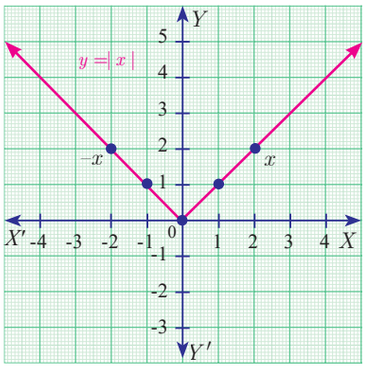


Fig. 1.46

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